

Persistence

Lecture 9 - CMSE 890

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Goals

Goals for today:

- Persistent homology

Need to do HW:

- Eugene
- Edem
- Jannik

Section 1

Last time

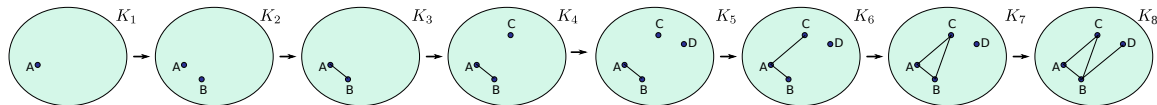
Last time: Simplicial Complex Filtration

Definition

A filtration $\mathcal{F} = \mathcal{F}(K)$ of a simplicial complex K is a nested sequence of its subcomplexes

$$\mathcal{F} : \emptyset = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = K.$$

$\mathcal{F}$ is called *simplex-wise* if $K_i \setminus K_{i-1}$ is empty or a single simplex.



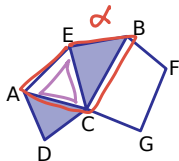
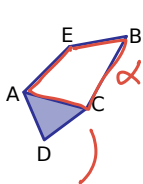
Betti curve

The p -th Betti curve is a function

$$\beta_p(\mathcal{F}) : \mathbb{R} \rightarrow \mathbb{Z}$$

$$i \mapsto \beta_p(K_i) = \text{rk}(H_p(K_i))$$

Induced map

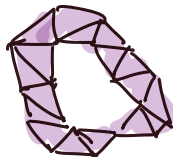


Given a simplicial map $f : K \rightarrow L$, the induced map on homology is defined by

$$\begin{aligned} f_* : H_p(K) &\rightarrow H_p(L) \\ [\alpha] &\mapsto [f_{\#}(\alpha)] \end{aligned}$$

$[\alpha]$

$$[\alpha] = [\underline{AE + EC + AC}]$$



Section 2

Persistence

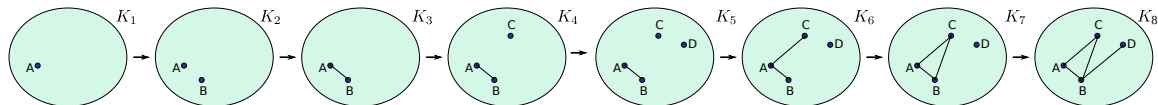
Persistence module

Given a filtration $\mathcal{F}$, the p -dimensional persistence module is

$$H_p(\mathcal{F}) : 0 = H_p(K_0) \rightarrow H_p(K_1) \rightarrow H_p(K_2) \cdots \rightarrow H_p(K_n) = H_p(K)$$

with maps $h_p^{i,j} : H_p(K_i) \rightarrow H_p(K_j)$ induced by inclusion.

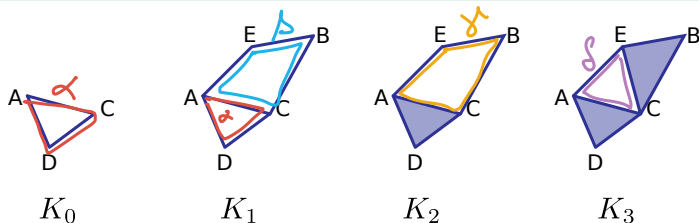
What's going on algebraically?



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

$$\begin{array}{cccccccc}
 [A] & \mapsto & [A] & \mapsto & [A] & \mapsto & [A] & \mapsto & [A] & \mapsto & [A] & \mapsto & [A] & \mapsto & [A] \\
 & & & & \swarrow & & & & \swarrow & & & & \swarrow & & \\
 & & & & [B] & & & & [C] & & & & [D] & & \\
 & & & & \swarrow & & & & \swarrow & & & & \swarrow & & \\
 & & & & [A] + [B] & & & & [C] & & & & [D] & & \\
 & & & & & & & & [C] & & & & [D] & & \\
 & & & & & & & & \swarrow & & & & \swarrow & & \\
 & & & & & & & & [D] & & & & [D] & & \\
 & & & & & & & & \swarrow & & & & \swarrow & & \\
 & & & & & & & & [D] & & & & [D] & &
 \end{array}$$

Another example

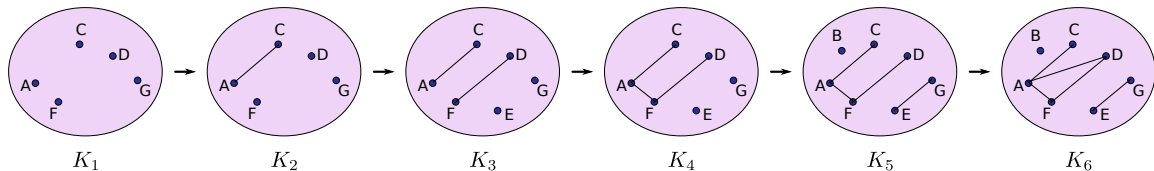


$$H_1(K_0) \xrightarrow{f_*} H_1(K_1) \xrightarrow{g_*} H_1(K_2) \xrightarrow{h_*} H_1(K_3)$$

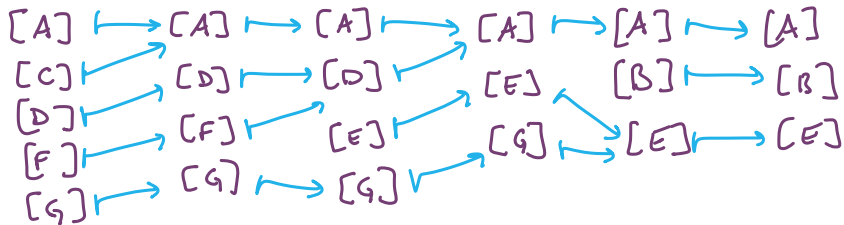
Diagram illustrating the mapping of homology classes $[\alpha], [\beta], [\gamma], [\delta]$ across the stages K_0, K_1, K_2, K_3 .

- $[\alpha]$ (red) maps to $[\alpha]$ (red) in $H_1(K_1)$.
- $[\beta]$ (blue) maps to $[\gamma]$ (yellow) in $H_1(K_2)$.
- $[\gamma]$ (yellow) maps to $[\delta]$ (purple) in $H_1(K_3)$.

Tryit: Figure out where the generators go



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6)$$



The p th persistent homology groups

$$H_p(K_0) \rightarrow H_p(K_1) \rightarrow H_p(K_2) \rightarrow \cdots \rightarrow H_p(K_i) \rightarrow \cdots \rightarrow H_p(K_j) \rightarrow \cdots \rightarrow H_p(K_n)$$

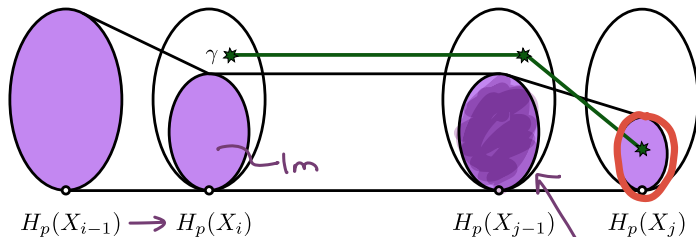
The p th persistent homology groups are the images induced by inclusion:

$$H_p^{i,j} = \text{Im}(H_p(K_i) \rightarrow H_p(K_j)).$$

The p th persistent Betti numbers are the ranks

$$\beta_p^{i,j} = \text{rank} H_p^{i,j}$$

Birth and Death



Birth ^[α]

A class $\gamma \in H_p(K_i)$ is born at K_i if it is not in $H_p^{i-1,i}$

Death

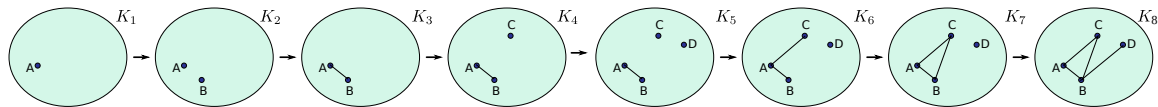
That class γ dies entering K_j if merges with an older class.^a

Specifically, if $h_p^{i,j-1}(\gamma) \notin H_p^{i-1,j-1}$ but $h_p^{i,j}(\gamma) \in H_p^{i-1,j}$.

^aWarning: The book's definition is different

When are generators born and die?

$$T: V \rightarrow W_{\geq \text{Im}(T)}$$



$$i-1=3$$

$$i=4$$

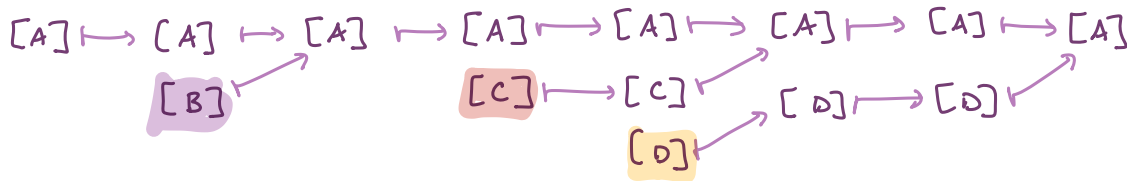
$$i-1=4$$

$$i=5$$

$$j=6$$

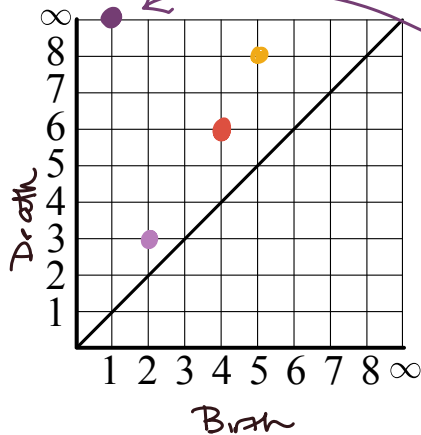
$$j=8$$

$$H_0(K_1) \rightarrow H_0(K_2) \rightarrow H_0(K_3) \rightarrow H_0(K_4) \rightarrow H_0(K_5) \rightarrow H_0(K_6) \rightarrow H_0(K_7) \rightarrow H_0(K_8)$$



Elder rule and persistence diagram

When two classes merge, the younger is the one that “dies.”

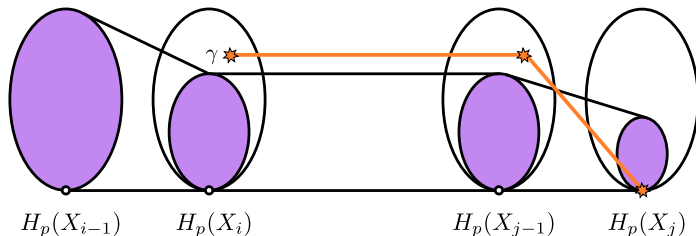


[A]
Born at 1
Never dies, lives forever
infinite class

Section 3

Dey Wang version of persistence classes

Birth and Death: Book version



Birth

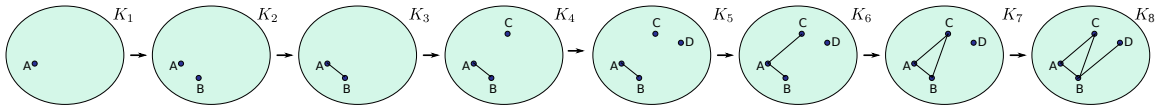
A class $\gamma \in H_p(K_i)$ is born at K_i if it is not in $H_p^{j-1,i}$

Death

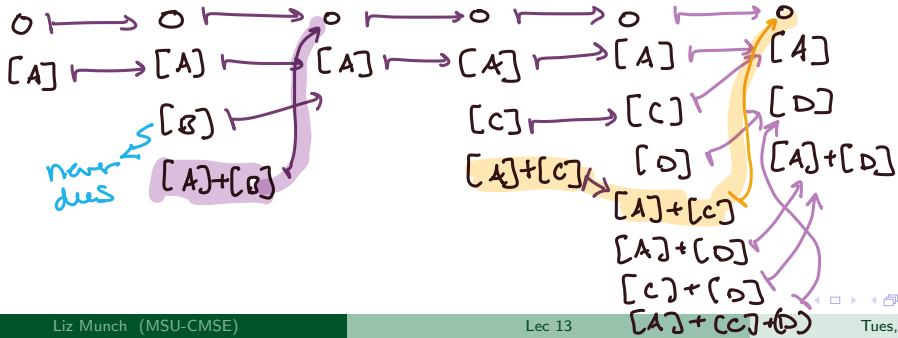
A class γ dies entering K_j if $\gamma \in H_p(X_{j-1})$ is not trivial, but $h_p^{j-1,j}(\gamma) = 0$.^a

^aWarning: In this version, not all classes die!

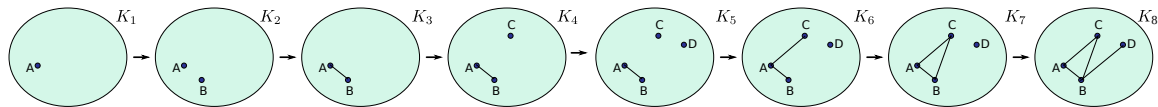
Lets check where (lots of but maybe not all) elements go



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$



Choosing a different basis



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

$$[A] \quad [A] \quad [A] \quad [A] \quad [A] \quad [A] \quad [A] \quad [A]$$

$$[A] + [B] \mapsto 0$$

$$[A] + [C] \mapsto 0$$

$$[A] + [D] \mapsto 0$$

$$\alpha = [A] + [C]$$

$$i_1 = 1 \quad c_1 = [A]$$

$$i_2 = 4 \quad c_4 = [C]$$

Book definition: which birth time pairs to the death time?

Let $[c]$ be a p -th homology class that dies entering X_j . Then, it is born at X_i if and only if there exists a sequence

$$i_1 \leq i_2 \leq \cdots \leq i_k = i$$

for some $k \geq 1$ so that

- There is a c_{i_ℓ} where $[c_{i_\ell}]$ is born at X_{i_ℓ} for every $\ell \in \{1, \dots, k\}$.
- $[c] = h_p^{i_1, j-1}([c_{i_1}]) + \cdots + h_p^{i_k, j-1}([c_{i_k}])$
- $i_k = i$ is the smallest possible value among any sequences have the above two properties.

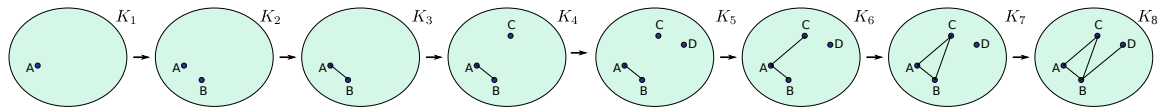
Counting classes

Persistence Module
↓

$$0 \rightarrow H_p(K_1) \rightarrow H_p(K_2) \rightarrow H_p(K_3) \rightarrow \cdots \rightarrow H_p(K_n) \rightarrow 0$$

- Attach 0 vector space at the end
- Associate $n + 1$ to $a_{n+1} = \infty$
- $\beta_p^{i,j} = \text{rank } H_p^{i,j} = \text{rank}(\text{Im}(H_p(K_i) \rightarrow H_p(K_j)))$
counts classes born **at or before** i and dying **after** j

Same example again



$$H_0(K_1) \longrightarrow H_0(K_2) \longrightarrow H_0(K_3) \longrightarrow H_0(K_4) \longrightarrow H_0(K_5) \longrightarrow H_0(K_6) \longrightarrow H_0(K_7) \longrightarrow H_0(K_8)$$

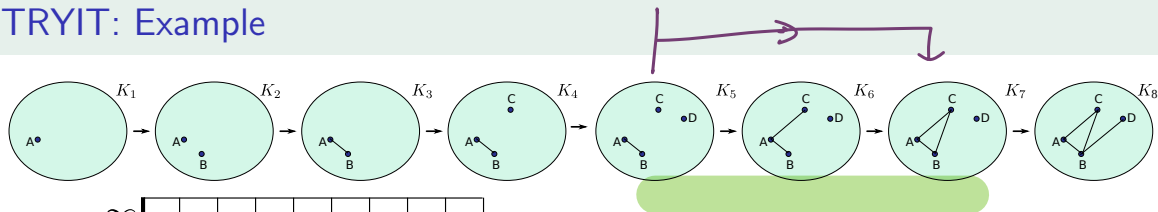
$$[A] \longmapsto [A] \longmapsto [A] \longmapsto [A] \longmapsto [A] \longmapsto [A] \longmapsto [A] \longmapsto [A]$$

$$[A + B] \longmapsto 0$$

$$[A + C] \mapsto [A + C] \longmapsto 0$$

$$[A + D] \mapsto [A + D] \mapsto [A + D] \longmapsto 0$$

TRYIT: Example



j

∞									
8	1	1	1	1	1	1			
7	1	1	1	1	2	2			
6	1	1	1	1	2	2			
5	1	1	1	2	3	X			
4	1	1	1	2	X	X			
3	1	1	1	X	X	X			
2	1	1	X	X	X	X			
1	1	X	X	X	X	X			
	1	2	3	4	5	6	7	8	∞

i

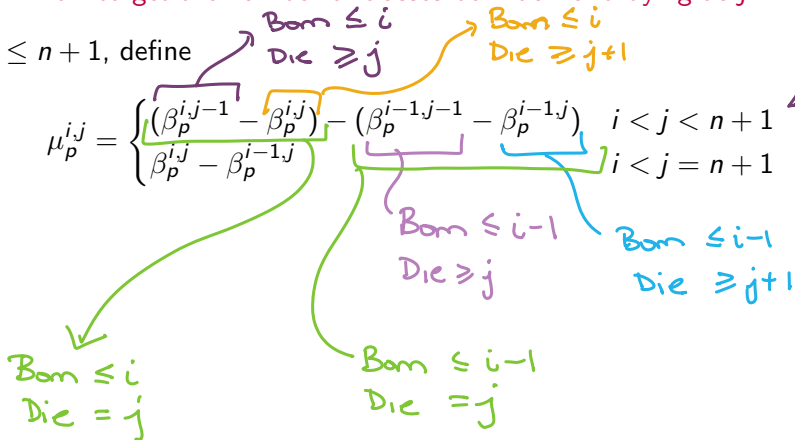
Determine $\beta_0^{i,j}$ for all pairs $i \leq j$ for this example

Persistence pairing function

How to get the number of classes born at i and dying at j ???

For $0 < i < j \leq n+1$, define

$$\mu_p^{i,j} = \begin{cases} (\beta_p^{i,j-1} - \beta_p^{i,j}) - (\beta_p^{i-1,j-1} - \beta_p^{i-1,j}) & i < j < n+1 \\ \beta_p^{i,j} - \beta_p^{i-1,j} & i < j = n+1 \end{cases}$$



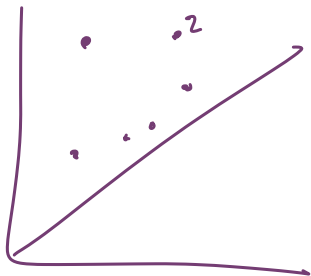
Persistence of a class

For $\mu_p^{i,j} \neq 0$, the persistence $Pers([c])$ of a class $[c]$ that is born at X_i and dies at X_j is defined as $Pers([c]) = a_j - a_i$. When $j = n + 1$ with $a_{n+1} = \infty$, $Pers([c]) = \infty$.

↳ The lifetime

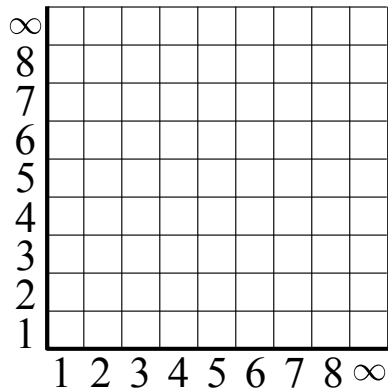
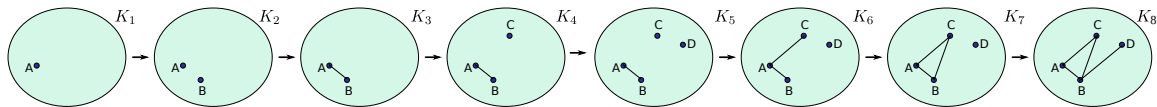
Persistence diagram

The persistence diagram $Dgm_p(\mathcal{F})$ (also written $Dgm_p(f)$) of a filtration induced by a function f is obtained by drawing a point (a_i, a_j) with non-zero multiplicity $\mu_p^{i,j}$ ($i < j$), on the extended plane where the points on the diagonal $\Delta = \{(x, x) \in \mathbb{R}^2\}$ are added with infinite multiplicity.



multiple things
Can be born and
die @ same time

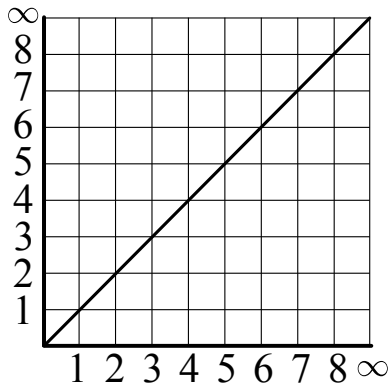
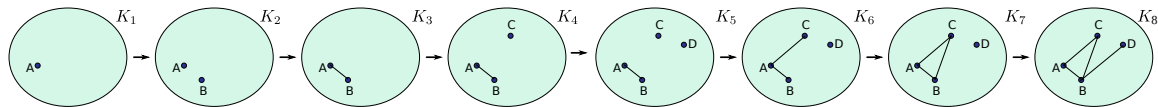
TRYIT: Example



Determine $\mu_0^{i,j}$ for all pairs $i < j$ for this example

$$\mu_p^{i,j} = \begin{cases} (\beta_p^{i,j-1} - \beta_p^{i,j}) - (\beta_p^{i-1,j-1} - \beta_p^{i-1,j}) & i < j < n+1 \\ \beta_p^{i,j} - \beta_p^{i-1,j} & i < j = n+1 \end{cases}$$

TRYIT: Persistence diagram for this example

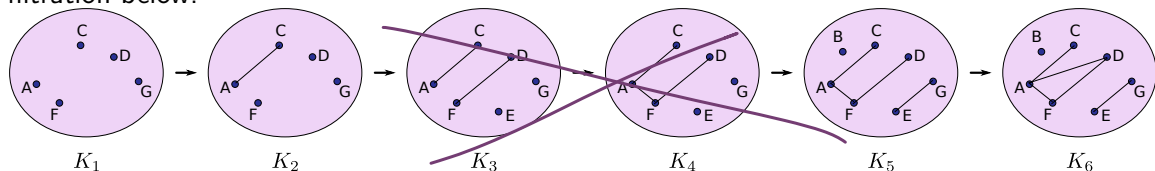


Plot the 0-dimensional persistence diagram

Homework

Jannik

Determine the $\beta_0^{i,j}$ table, the $\mu_0^{i,j}$ table, and the 0-dimensional persistence diagram for the filtration below.



Gn version